

Element C: Presentation and Justification of Solution Design Requirements

Priorities:

The following list is written in descending order of importance.

1. Safety (High Priority)

- The toy must cause no physical harm to the leopard or zookeepers interacting with it.
- Safety is assessed by ensuring no sharp edges or break-off pieces that could puncture skin or be ingested.
- This aligns with best practices in field enrichment, which practice risk assessment to prevent injury before enrichment is placed in an exhibit¹
- Preventing injury supports the primary stakeholder's goal of improving quality of life; any injury would reduce the animal's well-being or a keeper's ability to care for the leopard.

2. Cleanability (High Priority)

- The toy must be able to be fully cleaned within 3 minutes using high-pressure hosing without loss of function. Fully cleaned meaning no visible food residue, no detectable odor, and all crevices cleaned of debris
- This requires a design with accessible surfaces and minimal crevices, allowing staff to remove debris quickly.
- The 3-minute limit is based on direct communication with zoo personnel regarding daily cleaning routines.
- Effective cleaning prevents bacterial growth and transfer, which is critical in animal care environments.

3. Difficulty to Solve (High Priority)

- The puzzle must require at least two distinct steps to complete before obtaining the reward.
- This measurement stems from stakeholder input urging towards more time spent on puzzles, since more time spent on puzzles means more time enriched
- Ensuring multiple steps engages the leopard's problem-solving abilities and supports psychological stimulation goals.

4. Durability (High Priority)

- All interactive parts of the toy must withstand the leopard's maximum bite force without deformation.
- For this project, deformation is defined as a >10% alteration in any dimension or a loss of intended function.
- The toy must resist a bite force of at least 310 PSI, which is within the established estimated range of a leopard's bite force.²
- This requirement ensures the toy remains functional over repeated use and prevents hazardous breakage during play.

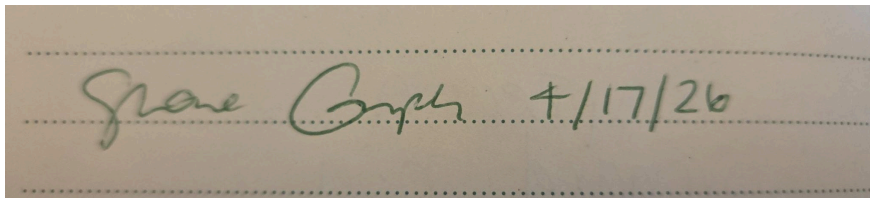
5. Longevity (High Priority)

- The toy must maintain structural integrity when used 1–2 times per week for multiple hours per session.
- Deformation again is defined as >10% change or loss of functionality.
- Longevity prevents frequent replacement, reducing costs and time burdens on zookeepers

6. Mobility (High Priority)

- The toy must be portable by a team of two zookeepers. To meet this requirement, the toy's loaded weight must be ≤ 40 pounds, and its volume ≤ 5 cubic feet.
- These limits are based on direct communication with zoo staff about safe handling and ergonomics.
- Portable design ensures the toy can be routinely introduced into exhibit

ZOOKEEPER AGREEMENT:

A photograph of a piece of lined paper with a handwritten signature "Shane Byles" and the date "4/17/26" in green ink.

Citations:

- Wild Enrichment Safety Overview. Wild Enrichment, wildenrichment.com/wp-content/uploads/2020/05/Felid-Enrichment-Overview.pdf. Accessed 2026.
- "What Is the Bite Force of a Leopard?" Biology Insights, 16 Dec. 2025, biologyinsights.com/what-is-the-bite-force-of-a-leopard/. Accessed March 2026.